Palmer Penguins - Faceting and Logistic Regression

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Palmer Penguins

Load the data

```
library(dplyr)
library(palmerpenguins)
library(ggplot2)

# Load the penguins dataset
penguins
```

```
## # A tibble: 344 x 8
##    species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##    <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
##  1 Adelie  Torgersen           39.1          18.7               181        3750
##  2 Adelie  Torgersen           39.5          17.4               186        3800
##  3 Adelie  Torgersen           40.3          18                 195        3250
##  4 Adelie  Torgersen           NA            NA                  NA          NA
##  5 Adelie  Torgersen           36.7          19.3               193        3450
##  6 Adelie  Torgersen           39.3          20.6               190        3650
##  7 Adelie  Torgersen           38.9          17.8               181        3625
##  8 Adelie  Torgersen           39.2          19.6               195        4675
##  9 Adelie  Torgersen           34.1          18.1               193        3475
## 10 Adelie  Torgersen           42            20.2               190        4250
## # i 334 more rows
## # i 2 more variables: sex <fct>, year <int>
```

Exploring the data

First, we can look at how many penguins there are from each species.

```
dplyr::count(penguins, species)
```

```
## # A tibble: 3 x 2
##   species       n
```

```
##   <fct>     <int>
## 1 Adelie      152
## 2 Chinstrap    68
## 3 Gentoo      124
```

There are three species of penguins: Adelie, Chinstrap, and Gentoo.

We can also look at the number of penguins on each island.

```
dplyr::count(penguins, island)
```

```
## # A tibble: 3 x 2
##   island        n
##   <fct>     <int>
## 1 Biscoe      168
## 2 Dream       124
## 3 Torgersen    52
```

There are three islands: Biscoe, Dream, and Torgersen.

We can examine the distribution of species across the islands.

```
penguins |> count(island, species)
```

```
## # A tibble: 5 x 3
##   island    species       n
##   <fct>     <fct>     <int>
## 1 Biscoe    Adelie       44
## 2 Biscoe    Gentoo      124
## 3 Dream     Adelie       56
## 4 Dream     Chinstrap    68
## 5 Torgersen Adelie       52
```

The different penguin species are found on different islands.

We can also look at the number of penguins by sex.

```
dplyr::count(penguins, sex)
```

```
## # A tibble: 3 x 2
##   sex        n
##   <fct>  <int>
## 1 female   165
## 2 male     168
## 3 <NA>      11
```

There are 165 female penguins and 168 male penguins. Some observations have missing values for sex.

Plot

We can examine the relationship between flipper length and bill length, while using color to distinguish between sexes.

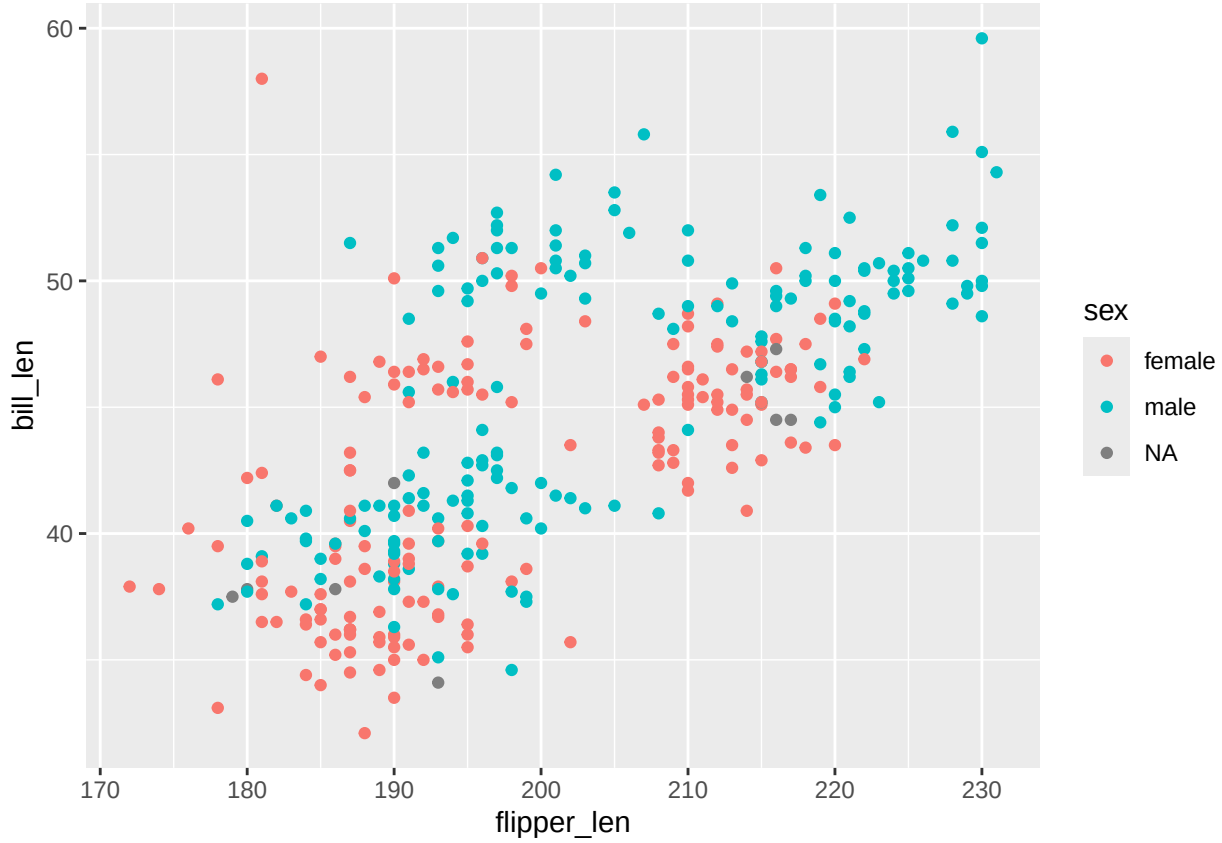
The original `palmerpenguins` dataset uses the variable names `flipper_length_mm` and `bill_length_mm`. For convenience, these are renamed below.

```
penguins <- penguins |>
  rename(
    flipper_len = flipper_length_mm,
    bill_len = bill_length_mm,
    body_mass = body_mass_g
  )
```

We can now create a scatter plot.

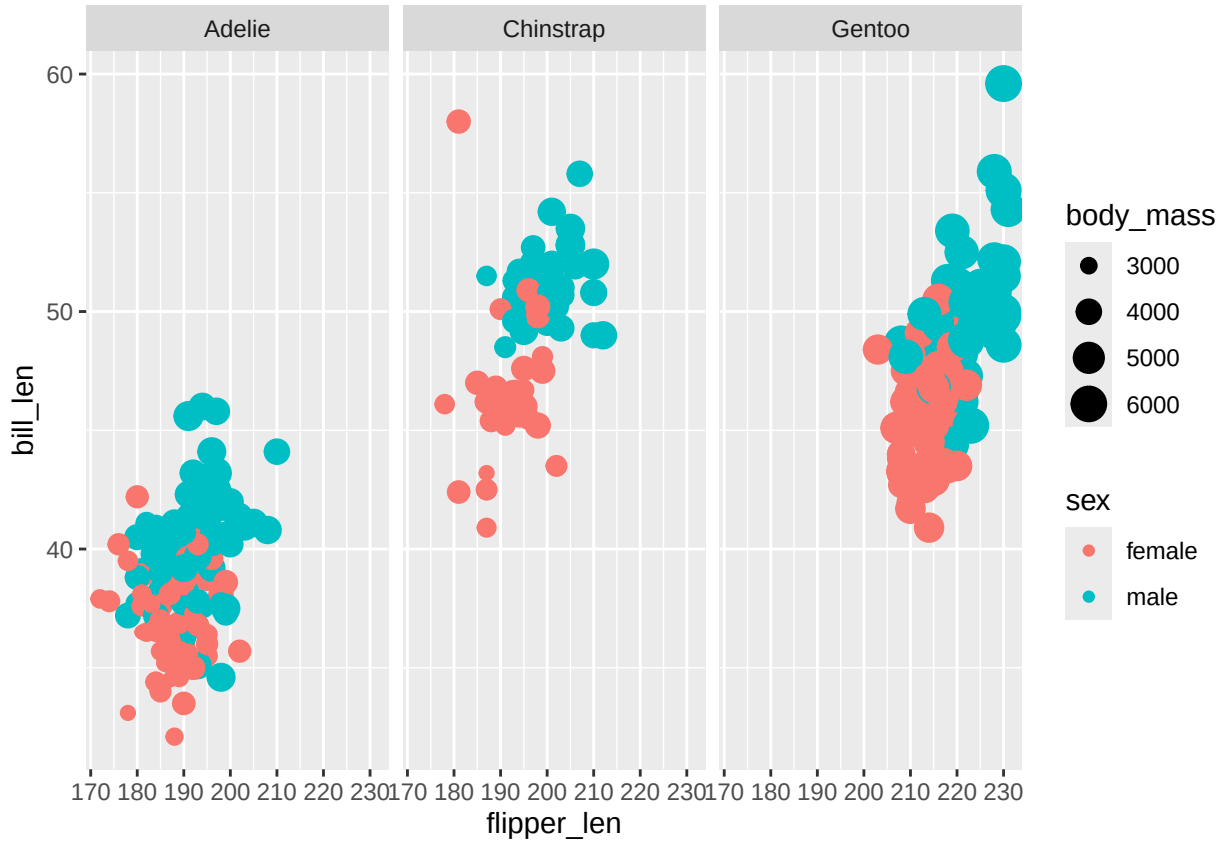
```
penguins |>
  ggplot(aes(flipper_len, bill_len, color = sex)) +
  geom_point()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Because there are some missing values for sex, we remove these observations before creating a more detailed plot. We also use `facet_wrap()` to create a separate plot for each species.

```
penguins |>
  filter(!is.na(sex)) |>
  ggplot(aes(flipper_len, bill_len, color = sex, size = body_mass)) +
  geom_point() +
  facet_wrap(~species)
```



For all three species of penguins, female penguins generally have smaller flippers and shorter bills than male penguins.

Logistic Regression

We can use logistic regression to investigate whether bill length, flipper length, and body mass can predict penguin sex.

First, we remove observations with missing values in any of the variables used in the model.

```
penguins_clean <- na.omit(penguins[, c(
  "sex", "bill_len", "flipper_len", "body_mass"
)])
```

We then fit a logistic regression model using `sex` as the response variable.

```
model <- glm(
  sex ~ bill_len + flipper_len + body_mass,
  data = penguins_clean,
```

```r
  family = binomial
)

summary(model)
```

```
##
## Call:
## glm(formula = sex ~ bill_len + flipper_len + body_mass, family = binomial,
##     data = penguins_clean)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  8.2569324  2.7636399   2.988  0.00281 **
## bill_len     0.1458945  0.0332279   4.391 1.13e-05 ***
## flipper_len -0.1348614  0.0234043  -5.762 8.30e-09 ***
## body_mass    0.0029604  0.0004188   7.069 1.56e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 461.61  on 332  degrees of freedom
## Residual deviance: 350.73  on 329  degrees of freedom
## AIC: 358.73
##
## Number of Fisher Scoring iterations: 4
```

Interpretation

The logistic regression results show that bill length and body mass were positively associated with the odds of being male, whereas flipper length was negatively associated with the odds of being male. All three predictors were statistically significant ($p < .001$).

The positive coefficient for bill length means that, holding the other variables constant, larger bill length is associated with higher odds of being male.

The negative coefficient for flipper length means that, holding the other variables constant, larger flipper length is associated with lower odds of being male.

The positive coefficient for body mass means that, holding the other variables constant, greater body mass is associated with higher odds of being male.

Overall, the results suggest that bill length, flipper length, and body mass are significant predictors of penguin sex.